

Wangzhi Zhan

Ph.D. Student, Computer Science, Virginia Tech

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Profile

Ph.D. student in Computer Science at Virginia Tech, advised by Prof. Dawei Zhou. My research focuses on generative AI, multimodal learning, and human-AI systems for scientific discovery. I build unified models, benchmarks, and interactive systems for controllable generation, evaluation, and exploration, with strong interests in foundation models, agentic AI, and scalable machine learning systems.

Education

Virginia Tech , Ph.D. in Computer Science, Blacksburg, VA, USA	Fall 2024 – Present
Advisor: Prof. Dawei Zhou	
China Institute of Atomic Energy (CIAE) , M.S., China	Fall 2020 – Spring 2023
Tsinghua University , B.S., Beijing, China	Fall 2016 – Spring 2020

Skills

Programming: Python, Git, LaTeX

ML Areas: LLMs, diffusion models, multimodal learning, geometric deep learning, AI for Science, human-in-the-loop AI systems

Research & Engineering: model training, benchmarking, evaluation pipeline design, interactive demo development, scientific data modeling

Research Experience

Research Assistant, Virginia Tech Fall 2024 – Present

- Develop generative and predictive AI models for scientific discovery, with a focus on controllable structure generation, property prediction, and reliable evaluation.
- Build unified frameworks for 3D metamaterial design, including diffusion-based generation, condition-aware prediction, and benchmark development.
- Design interactive human-AI systems that support scientific ideation, exploration, and automated design workflows.
- Construct datasets, evaluation metrics, and experimental pipelines for open-ended scientific learning and out-of-distribution generalization.

Selected Projects

- **UniMate**
 - Built a unified model for mechanical metamaterial generation, property prediction, and condition confirmation.
 - Developed an end-to-end framework for controllable 3D structure design and evaluation.
 - Code: github.com/wzhan24/UniMate
- **MetamatBench**
 - Built a benchmark integrating heterogeneous datasets, computational tools, and visual interfaces for metamaterial discovery.
 - Supported robust model comparison and interactive exploration for AI-driven scientific design.
 - Code: github.com/cjpcool/Metamaterial-Benchmark
- **MetaScientist**
 - Contributed to a human-AI collaborative system for automated mechanical metamaterial design.
 - Helped connect AI generation, scientific reasoning, and user-facing interaction in a unified design workflow.

Selected Publications

- **UniMate: A Unified Model for Mechanical Metamaterial Generation, Property Prediction, and Condition Confirmation.** Wangzhi Zhan, Jianpeng Chen, Dongqi Fu, Dawei Zhou. *ICML*, 2025.
- **MetamatBench: Integrating Heterogeneous Data, Computational Tools, and Visual Interface for Metamaterial Discovery.** Jianpeng Chen, Wangzhi Zhan, Haohui Wang, Zian Jia, et al. *KDD*, 2025.
- **MetaScientist: A Human-AI Synergistic Framework for Automated Mechanical Metamaterial Design.** Jingyuan Qi, Zian Jia, Minqian Liu, Wangzhi Zhan, et al. *NAACL Demo*, 2025.
- **The End of Trial-and-Error: A Vision for Generative Intelligence in Metamaterial Design.** Adithya Kulkarni, Haohui Wang, Wangzhi Zhan, Jianpeng Chen, Dawei Zhou. *ICDMW*, 2025.
- **MetaRef: A Generalizable Physics-Aware Refinement Framework for Metamaterial Design.** Alexander Lu, Wangzhi Zhan, Jianpeng Chen, Dawei Zhou. *ICDMW*, 2025.

Selected Manuscripts Under Review

- **ReGuiLD: Guided Latent Diffusion in a Regulated Latent Space for Metamaterial Structure Exploration.** Wangzhi Zhan, Jianpeng Chen, Dawei Zhou. Submitted to *ICML*, 2026.
- **DuetDA: Decomposed and Dynamic Data Attribution with Model-State Gating for Accelerated Scientific Endeavors.** Jianpeng Chen, Wangzhi Zhan, Haohui Wang, Dongqi Fu, Dawei Zhou. Submitted to *KDD*, 2026.

Academic Service

Reviewer for ICLR 2026, ICML 2026, and *ACM Computing Surveys*.